Multipurpose Service Robot

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The authors gratefully acknowledge Dr. Nilima Warke for proposing the initial idea for this project and for her guidance during its development.

***Abstract*—Manual collection of scattered waste is repetitive and may expose sanitation personnel to unhygienic and potentially unsafe working conditions. This paper presents the design and ongoing development of a prototype multipurpose mobile service robot for dry-waste detection and collection. The platform combines a four-wheel mobile base, a servo-actuated robotic arm and gripper, embedded control, environmental sensing and a camera-based object-recognition pipeline. The proposed operating sequence is patrol, detect, approach, align, pick, dispose and repeat, with a safe-stop response intended to override normal motion when an obstacle or fault is detected. Chassis movement and arm range of motion have been reported as functionally checked in the project validation material, whereas vision integration with the mobile base and manipulator, obstacle-aware navigation and the complete autonomous cycle remain pending. Sample vision-interface outputs are treated as subsystem demonstrations only, not as dataset-level accuracy. No unmeasured end-to-end performance is claimed; the contribution is a documented prototype design, an evidence-bounded implementation record and a repeatable evaluation protocol.**

***Index Terms*—Mobile robot, waste collection, computer vision, YOLO, robotic manipulation, dry-waste segregation, embedded control, service robotics.**

I. INTRODUCTION

Municipal and institutional environments generate a continuous stream of dry waste such as paper, cardboard, plastic containers and metal packaging. In many settings, the first-stage collection task remains dependent on manual inspection and handling, which is repetitive and may expose workers to sharp edges, contaminated surfaces and physically demanding movements. A compact robotic platform can assist with selected tasks by combining mobility, sensing and manipulation.

A waste-collection robot must solve a sequence of coupled problems rather than a single classification task: it must move through a bounded area, detect an object, estimate whether the object is reachable, position its base, operate a gripper, transfer the object and verify that the task has been completed. A failure in any stage can prevent successful collection even when the vision model produces the correct class, so the design problem is one of integration across mechanics, control, perception, power and safety.

The project is motivated by the need for a compact and comparatively low-cost prototype that can be expanded in stages, while keeping the difference between demonstrated subsystem behaviour and planned autonomous functions explicit. The objectives are: (1) develop a four-wheel mobile robotic platform; (2) provide a manipulator and gripper for lightweight dry-waste pickup; (3) incorporate an embedded controller and suitable motor/servo drive circuitry; (4) plan camera-based object detection and environmental sensing; (5) define a state-based operating sequence that links detection to action; and (6) establish measurable tests for detection, mobility, manipulation, safety and complete-cycle operation.

The scope is limited to prototype-scale indoor testing on a level surface and lightweight dry-waste samples; the project is a controlled prototype, not a finished outdoor street-cleaning product. The paper contributes a system description, a literature comparison, a staged implementation record, and an evaluation framework that assigns no performance values to tests not yet completed.

II. REVIEW OF LITERATURE

Lahoti et al. describe a multi-class waste-segregation system using YOLOv5 and a robotic arm; their controlled test setup reports successful segregation in eight out of ten attempts, illustrating the feasibility of connecting object recognition with physical sorting [1]. However, a fixed or controlled camera arrangement differs from a mobile robot, where object distance, viewpoint, lighting and base position can change between observations. Harada et al. investigate trash detection suitable for mobile robots using an improved YOLO approach [2], and SHIFT-T presents a low-cost mobile trash collection and sorting platform using YOLOv5-based perception, a camera, ultrasonic sensing and a mobile manipulator [3]. These studies show that mobility, localisation and manipulation must be coordinated; a detection label alone does not specify a safe grasp pose. Localisation research stresses that object position and geometry must be estimated before grasping, since a bounding box is in camera coordinates while the manipulator works in its own frame [4].

Lu and Chen review computer-vision research for solid-waste sorting, identify the transition from handcrafted features toward deep learning, and note that many studies rely on simplified environments and privately collected datasets, which can limit generalisation [5]. Lin et al. combine YOLO-based detection with a delta robot and report that mechanical efficiency is a major factor in overall sorting performance [6], and Ha et al. describe a YOLO-based sorting system with a multi-degree-of-freedom arm [7]. Table I is a qualitative synthesis, not a controlled comparison of measured accuracies; success values from different papers should not be compared directly because datasets, conditions and protocols differ.

TABLE I
COMPARISON OF REPRESENTATIVE WASTE-SORTING RESEARCH

Study	Approach	Contribution	Limitation for this project
Lahoti et al. [1]	YOLOv5 + arm	Multi-class dry-waste sorting	Controlled camera setup; mobile navigation differs
Harada et al. [2]	Improved YOLO	Trash detection for mobile robots	Detection does not itself execute grasp/disposal
SHIFT-T [3]	YOLOv5, camera, sonar, mobile arm	Mobile trash detection and sorting	Requires coordinated localisation and manipulation
Vision localisation [4]	Visual geometry	Object localisation for grasping	Needs camera/robot calibration
Lu & Chen [5]	Critical review	Maps CV methods and dataset challenges	Highlights simplified test environments
Lin et al. [6]	YOLO + delta robot	Automated recyclable sorting	Mechanical efficiency affects throughput
Ha et al. [7]	YOLOv4 + arm	Intelligent waste sorting control	Offline dataset testing differs from mobile use

Waste detection and robotic sorting are active research areas; the practical gap addressed here is the staged integration of a compact mobile base, an articulated arm, embedded control, environmental sensing and vision-based target selection. The project does not claim a new YOLO architecture or a new manipulation algorithm; its focus is the system-level engineering needed to connect the modules and a test protocol that can expose failures in each stage.

III. METHODOLOGY

A. Requirements and Staged Approach

The robot is expected to initialise its controller and sensors, move within a defined test boundary, observe the environment, detect a candidate waste item, determine whether the item is reachable, position the base, execute a pickup sequence, transfer the item to a collection location and return to patrol. It should stop when an obstacle or abnormal condition is detected, and the operator must be able to stop the system. The initial environment is a level indoor floor with adequate lighting, a bounded test region and lightweight dry-waste samples; outdoor operation is outside the validated scope. Work proceeds in six stages: (1) inspect and assemble the mobile chassis; (2) verify drive motors and basic movement; (3) calibrate the arm joints and gripper; (4) validate camera-based detection on a stationary setup; (5) connect detection, target positioning and actuator commands; and (6) evaluate the integrated cycle using repeated trials. This prevents a working motor test or a successful sample classification from being reported as complete autonomy.

B. System Architecture and Control

The architecture contains sensing, perception, decision, actuation and verification blocks (Fig. 1). A camera frame is passed to the object detector, which returns a class label, confidence score and image-space bounding box. Target selection rejects detections below the threshold or outside the valid region; a valid target is mapped to a robot-relative position, the base is positioned so that the target lies within the arm workspace, and only then does the arm sequence begin.

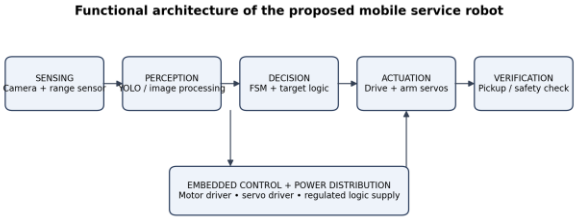


Fig. 1. Functional architecture of the proposed mobile service robot.

Task organisation uses a finite-state machine with the conceptual states IDLE, INITIALISE, PATROL, DETECT, APPROACH, ALIGN, PICK, DISPOSE, VERIFY, RESUME and SAFE STOP, with event-driven transitions. Obstacle detection and emergency stop have priority over ordinary states: if a range sensor reports an obstruction inside the configured safety distance, drive commands are inhibited and the system enters SAFE STOP; if the object is lost during approach, the controller stops and returns to DETECT; and if pickup verification fails, the arm returns to a safe pose and the attempt is logged as unsuccessful. This sequence is a proposed workflow; the current project evidence does not establish that every transition is already implemented and operating autonomously.

C. Hardware Design

The prototype uses a four-wheel chassis with a flat upper deck that supports the battery pack, controller board, motor-driver circuitry, wiring and arm assembly. Geared DC motors and a motor-driver stage provide forward, reverse and turning commands. In a differential-drive approximation, the forward velocity is $v = r(\omega_R + \omega_L)/2$ and the angular velocity is $\Omega = r(\omega_R - \omega_L)/b$, where r is the effective wheel radius, ω_L and ω_R are the left and right angular velocities and b is the track width. The left-side and right-side motor groups must be calibrated as paired drives on the actual test surface. A servo-actuated articulated arm with a gripper handles lightweight objects; the base remains stationary during pickup because the extended arm shifts the centre of gravity.

The embedded controller handles low-level actuation and sensor interfacing. The review material identifies an ESP32-based control approach, while the supplied prototype photograph shows a development board with an Arduino-style layout; no unverified exact board revision is therefore assigned. The planning material lists an L298N motor driver and a PCA9685 servo driver as intended components, and the final installed configuration must be confirmed. A Li-ion battery pack supplies the motors, servos and logic through regulation, arranged to reduce resets caused by servo current transients. The proposed sensing arrangement is a camera for object recognition and an RPLIDAR / range sensor for obstacle awareness, whose integration and validation are pending.

D. Software and Vision Pipeline

The software is divided into perception, decision-making, low-level control and monitoring. Camera frames are preprocessed and passed to a YOLO detector, a single-stage family that predicts classes and bounding regions in one pass; its confidence score is a per-frame model output, not the probability that the system will collect the object. Target selection filters detections by confidence, class eligibility, bounding-box size and proximity to the valid pickup region, and rejects detections that are partly outside the image, heavily occluded or beyond the calibrated workspace. Image coordinates are mapped to robot-relative coordinates using camera intrinsics and a camera-to-robot transform, or a planar calibration such as a homography; until this is completed and tested, bounding boxes are not treated as direct joint commands.

The vision process and embedded controller exchange validated commands whose acknowledgements distinguish receipt (RECEIVED) from completion (COMPLETED or FAILED), with timeouts so that a pickup is not assumed successful merely because a message was sent. The confidence threshold should be selected on a validation set and fixed before the final test set is evaluated, and each trial is logged (state, class, confidence, bounding box, sensor readings, commanded action, completion flag and failure code).

E. Evaluation Protocol

Detection is evaluated against manually annotated held-out images using precision, TP/(TP + FP), recall, TP/(TP + FN), and F1-score, 2 × Precision × Recall/(Precision + Recall), counted per class at a stated intersection-over-union threshold. Manipulation is measured by pickup success (successful grasp-and-transfer attempts / total pickup attempts × 100), mobility by navigation success (completed route trials / total route trials × 100) and obstacle response, and the system by cycle time and end-to-end success (complete detect-to-dispose trials / total full-cycle trials × 100). A complete-cycle success is counted only when detection, approach, alignment, pickup, transfer, disposal verification and return to the intended state all succeed in one trial. Runs with manual intervention are not counted as autonomous successes, failed trials remain in the dataset, and every percentage is reported with its trial count.

IV. RESULTS

Fig. 2 shows the physical prototype: a four-wheel chassis with yellow wheels, a wooden upper deck, a battery pack, a controller board, motor-driver circuitry, wiring, a mounted articulated arm with a gripper, and a front sensing module. The photograph documents the physical arrangement only; it does not demonstrate autonomous navigation or a complete vision-to-pickup cycle.

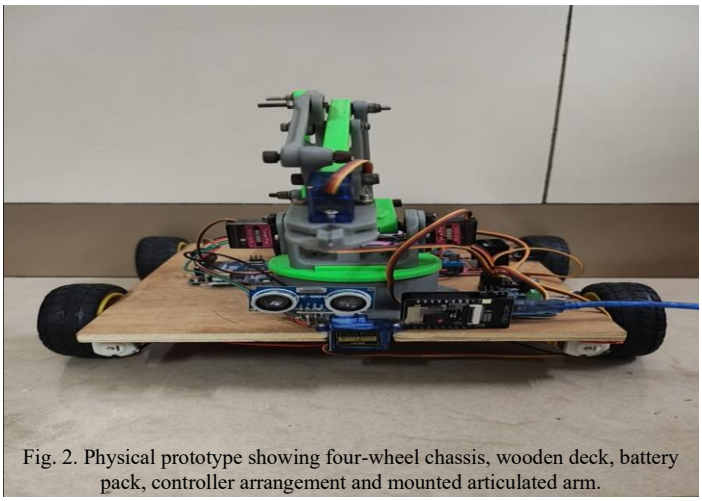


Fig. 2. Physical prototype showing four-wheel chassis, wooden deck, battery pack, controller arrangement and mounted articulated arm.

Table II summarises the implementation status. The project poster records the four-wheel chassis mobility tests (forward, reverse and turning) and the arm range-of-motion / pick-and-place reach across joints as verified, and records vision integration with the arm and navigation, obstacle-aware navigation and full autonomous patrol-detect-collect validation as pending.

TABLE II
IMPLEMENTATION STATUS BASED ON PROJECT EVIDENCE

Function	Evidence available	Current status
Four-wheel chassis	Physical prototype photograph; project poster validation	Assembled; forward/reverse/turn reported verified
Arm and gripper	Mounted arm visible; project poster validation	Range-of-motion / reach reported verified
Controller and power	Controller board and battery visible in photograph	Arrangement present; final wiring and under-load measurements need documentation
Vision sample output	Sample interface screenshots labelled plastic and metal	Sample outputs available; not dataset-level evaluation
YOLO-to-arm/navigation link	Project poster and review list integration as pending	YOLO is linked to the motors of the robot; linking to the arm will need time.
Obstacle-aware navigation	Project review lists navigation and obstacle avoidance as pending	Pending
Full autonomous cycle	Project poster lists full end-to-end validation as pending	Pending

Sample vision-interface outputs labelled plastic and metal are available. They demonstrate classification behaviour for the shown images only; the size and composition of a held-out test set, class-wise confusion counts, precision, recall and F1-score are not available, so no numerical detection accuracy is reported. Consequently, no measured collection success, navigation success, obstacle-avoidance rate, battery runtime or detection accuracy is claimed. The project poster lists target values of at least 80% detection accuracy and at least 70% pick-and-place success, both recorded as testing in progress; these are acceptance targets, not achieved results.

V. LIMITATIONS

The prototype is a compact educational platform intended for lightweight dry waste. Payload capacity, arm reach, servo torque, chassis stiffness and centre-of-gravity changes constrain its operating envelope; the wooden deck may flex under repeated loading, and the platform should not be described as suitable for outdoor operation until tested on representative surfaces. Camera performance can vary with lighting, reflections, clutter, occlusion and object orientation, and a model trained on a limited dataset may not generalise to unfamiliar waste items. The camera-to-robot coordinate transform, detector-to-controller command interface and pickup verification logic are critical integration points; if any is missing or unreliable, the complete autonomous cycle will fail even if individual modules operate. Testing should use low speed, lightweight non-hazardous objects and direct supervision; the system is not intended for medical waste, hazardous materials or unknown substances. The proposed system was initially envisioned to use a Raspberry Pi, LiDAR sensor, and ROS framework for more accurate autonomous navigation and improved robotic arm-based object detection and pickup. However, due to the high implementation cost, these components were excluded from the current prototype. Nevertheless, they can be integrated in future versions to enhance navigation accuracy, obstacle avoidance, and autonomous pick-and-place capabilities..

VI. CONCLUSION

This paper has presented the design and ongoing development of a multipurpose mobile service robot for vision-guided dry-waste collection, combining a four-wheel mobile base, an articulated servo-operated arm and gripper, embedded control, environmental sensing and a planned camera-based detection pipeline. The project validation material reports checks of chassis movement and arm range of motion, but the complete autonomous patrol-detect-approach-pick-dispose cycle has not yet been validated. The next steps are to document controller wiring and power distribution, integrate camera inference, establish camera-to-robot calibration, implement obstacle-aware navigation and connect the detector to the manipulation sequence, followed by repeated indoor trials for each waste class. The paper therefore avoids unsupported performance claims; its contribution is a structured design and implementation record and a measurable validation plan.

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